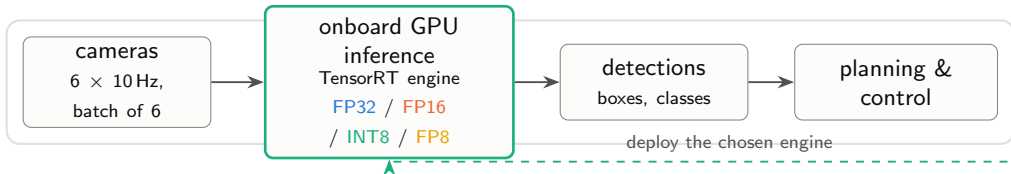


Vehicle (synchronised camera frames)



Offline PEP-AI validation

per precision
accuracy, head-input
deviation, critical errors
per image $K(p)$

measured
energy per image $E_{\text{img}}(p)$,
latency, jitter

AIBO decision layer

fleet size, operating hours, price c_{el} ,
grid intensity (g CO₂/kWh),
penalty λ per critical error

cost model
 $C(p, \lambda) = N E_{\text{img}} c_{\text{el}} + N K \lambda$
 $\text{CO}_2 \propto N E_{\text{img}}$

decision
precision per model,
break-even λ^*

